Lab 3: 5-stage Basic Pipeline

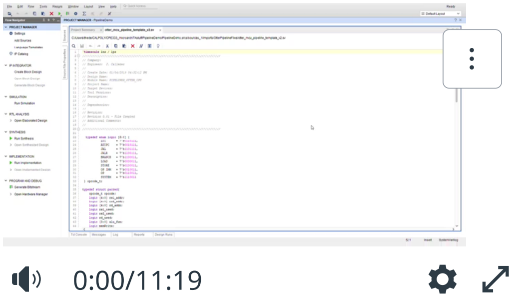
Start Assignment

- Due Jul 9 by 5pm
- Points 100
- Submitting a file upload

Basic Pipelined 5 Stages CPU

lab files: **OtterPipelineFiles.zip (https://canvas.calpoly.edu/courses/186931/files/21847886/download?wrap=1) (https://canvas.calpoly.edu/courses/186931/files/21847886/download?download_frd=1)**

Video on how to start lab:



On the report add the students names in your group and a Table of the **tasks each member** of the group performed.

To test the program since we are assuming no hazard (data or control) for testing we are going to be using a simple program:

addi x7, zero, 7

addi x8,zero, 40 (change this address to a Data memory address in your Otter)

addi x10,zero,10

nop

nop

or x11,x7,x10

nop

nop

sw x11,0(x8)

If you store the right value in memory, then we are done for this lab (show this on the waveform and highlight where this result is stored)

On the report :

1. add the students' names in your group and a Table of the **tasks each group member** performed.

2. contributions of every team member

3. All project files as a zip file or a github link where I (mpantoja314 has access)

3. Describe the design decisions the team took

4. Prove that it compiles, synthesize to the board by running the above code; you can make a video or demo to TA/ Instructor of your project compiling, simulating /creating correct waveform and on the Basys/Microchip board

Lab 4 Rubric

Criteria	Ratings	Pts
Summary of Project Design Detailed description of development process and design of pipeline.		20 pts
Verilog Code All new System Verilog code modules created or edited for this assignment.		40 pts
Verification Some kind of way to verify/demonstrate that the pipeline works as intended. Possibly a test program waveform.		40 pts
		Total Points: 100